

Date: 10-03-2026

Student Performance Feature Engineering using Python

For this task, I used a **Student Performance dataset** and applied different **feature engineering techniques using Python and the sklearn library**. The main goal of this task was to improve the dataset so that machine learning models can understand the data more easily.

The dataset contains information about students such as **gender, school type, study hours, attendance, motivation level, family income, tutoring sessions, and exam score**. These features help to understand the factors that affects student performance.

Encoding

In this task, I applied **encoding techniques** to convert categorical data into numerical format. Machine learning models cannot work with text data, so encoding is required.

For **nominal categorical variables** such as gender, school type, peer influence, and internet access, I used **OneHotEncoder from sklearn**. This technique converts each category into separate binary columns with values 0 and 1.

For **ordinal features** such as **motivation level, family income, and teacher quality**, I have used **mapping encoding**. This method assigns numbers to categories based on their order like Low = 0, Medium = 1, and High = 2. This helps to maintain the natural order of these categories.

Scaling

Scaling was applied to numerical features such as **hours studied, attendance, sleep hours, previous scores, and tutoring sessions**. I used **StandardScaler** from **sklearn** to standardize these values.

Standard scaling transforms the data so that the values have a **mean of 0 and a standard deviation of 1**. This helps ensure that features with larger values do not influence the learning process of the model.

Binning

I used **binning technique** to convert continuous numerical values into categories. For example, **study hours** were divided into categories such as **Low, Medium, and High**.

This technique helps to simplify the data and makes it easier to understand patterns in student behaviour.

Interaction Features

Interaction features were created by combining two features together.

I multiplied **hours studied with attendance** to create a new feature called **study effort**.

These features help to understand relationships between variables that may impact on the final exam score.